

Poster MATHS & K:

- Derivative estimate for used for second
- Integration ~~for~~ to move from acceleration to position
- Represents how through derivative & integration, you go back to forth.
- Used concept of limits to help represents dynamic desired values that moves.
- Add 2D graph
- Test runs when tuning & has infinite limits & discontinuity when the graphed showed "it in two different moments"

Note for Aircraft 442

$$T_{\text{Requirement}} = \frac{W}{C_L C_D} = \frac{W}{L/D}$$

- When given V_∞ you can find ^{the} stabilization lift coefficient by $C_L = \frac{W}{\frac{1}{2} \rho V_\infty^2 S}$

Then you calculate C_D by the formula we written & create the ratio $\frac{C_L}{C_D}$ to calculate thrust requirement

- The T_R can also be considered the addition of zero-lift thrust required & lift-induced thrust required

Power = $\frac{\text{energy}}{\text{time}}$, $P_R = T_R V_\infty$ to find what amount of power is required to keep aircraft stabilized

$$P_R = \sqrt{\frac{2W^3 C_D^2}{\rho S C_L^3}} \propto \frac{1}{C_L^{3/2} / C_D} \quad \text{the aerodynamic that holds as minimum power required is}$$

Need to read Ch. 6 Rate of Climb